

## Project Planning Phase

### Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

|                      |                                                    |
|----------------------|----------------------------------------------------|
| <b>Date</b>          | 13 July 2026                                       |
| <b>Team ID</b>       |                                                    |
| <b>Project Name</b>  | Voice-Based Concept Understanding Analyzer (VBCUA) |
| <b>Maximum Marks</b> | 5 Marks                                            |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create the product backlog and sprint schedule.

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                                       | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-1 | Audio Upload                  | USN-1             | As a student, I can upload a recorded audio file of my spoken explanation so that it can be evaluated.                                  | 3            | High     |              |
| Sprint-1 |                               | USN-2             | As a student, I want unsupported audio file types to be rejected so that I only upload a file the system can process.                   | 2            | Medium   |              |
| Sprint-1 | Speech-to-Text Transcription  | USN-3             | As a student, I want my audio automatically transcribed into text so that I can see what the system understood from my explanation.     | 5            | High     |              |
| Sprint-2 |                               | USN-4             | As a student, I want to be notified if transcription fails so that I know to re-upload my audio.                                        | 2            | Medium   |              |
| Sprint-1 | Reference Answer Management   | USN-11            | As an evaluator, I want the system to use a stored reference answer for comparison so that grading is based on a consistent standard.   | 3            | High     |              |
| Sprint-1 | Semantic Similarity Analysis  | USN-5             | As a student, I want my transcript compared against a reference answer so that my understanding can be measured objectively.            | 5            | High     |              |
| Sprint-1 | Concept Understanding Scoring | USN-6             | As a student, I want to receive a grade (Excellent / Good / Fair / Needs Improvement) so that I can quickly gauge my performance.       | 3            | High     |              |
| Sprint-1 |                               | USN-7             | As a student, I want written feedback explaining my grade so that I understand what to improve.                                         | 2            | Medium   |              |
| Sprint-1 | Results Display               | USN-8             | As a student, I want to view my transcript, score, and feedback together on one screen so that I can review my performance at a glance. | 3            | Medium   |              |

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                            | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|--------------|----------|--------------|
| Sprint-2 |                               | USN-12            | As an evaluator, I want each submission to show a clear grade tier so that I can quickly identify students who need support. | 2            | Medium   |              |
| Sprint-2 | PDF Report Generation         | USN-9             | As a student, I want a PDF report of my assessment generated automatically so that I can keep a record of my performance.    | 5            | Medium   |              |
| Sprint-2 | Report Download               | USN-10            | As a student, I want to download my PDF report so that I can save or share it.                                               | 2            | Medium   |              |

### Project Tracker, Velocity & Burndown Chart (4 Marks)

| Sprint   | Total Story Points | Duration | Sprint Start Date (Planned) | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-----------------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 26                 | 6 Days   | 01 Jun 2026                 | 06 Jun 2026               | 26                                              | 06 Jun 2026                  |
| Sprint-2 | 11                 | 6 Days   | 08 Jun 2026                 | 13 Jun 2026               | 11                                              | 13 Jun 2026                  |
|          |                    |          |                             |                           |                                                 |                              |
|          |                    |          |                             |                           |                                                 |                              |

#### Velocity:

Velocity is the amount of work (in story points) a team completes during a sprint. The average velocity (AV), expressed as story points completed per day, is calculated as:

Average Velocity (AV) = Total Story Points ÷ Sprint Duration (days)

For Sprint-1:  $AV = 26 \div 6 \approx 4.3$  story points per day. For Sprint-2:  $AV = 11 \div 6 \approx 1.8$  story points per day. The lower velocity in Sprint-2 reflects the smaller scope of the remaining stories (error handling, report generation, and download) compared to the core transcription and scoring pipeline built in Sprint-1.

#### Burndown Chart:

A burndown chart is a graphical representation of work remaining versus time. It is commonly used in agile software development methodologies such as Scrum to track a team's progress toward completing a sprint, though it can be applied to any project with measurable progress over time.

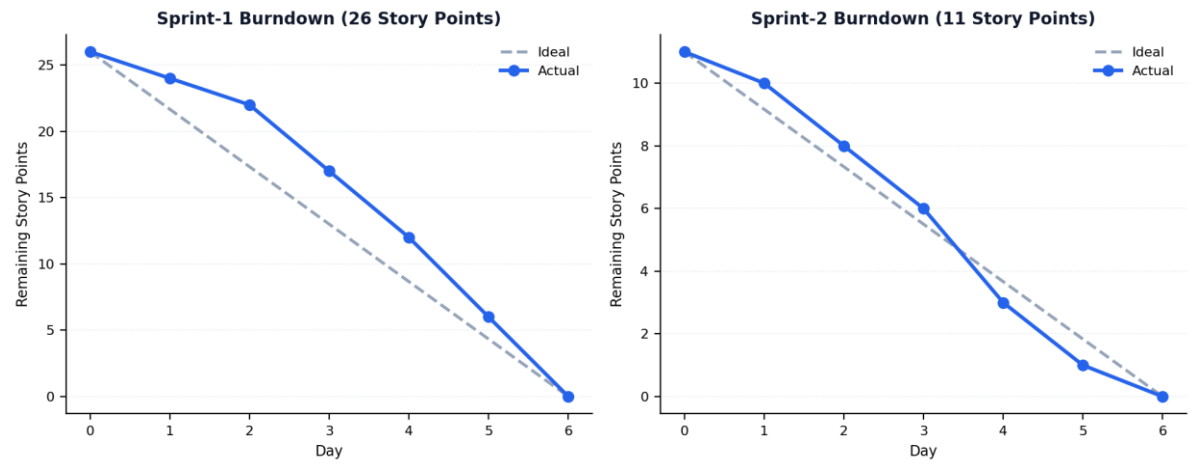


Figure 1: Sprint-1 and Sprint-2 Burndown Charts — VBCUA